

有方-科研竞赛培养方案

专为 24020 李欣阳 设计

课题名称

Decoding the Universe's Hidden Distortions: A Machine Learning Approach to Compress Astrophysical Systematics in Weak Lensing

宇宙隐藏扭曲的解码：压缩弱引力透镜中天体物理系统误差的机器学习方法

课题介绍

English

Weak gravitational lensing is one of the key observational probes for mapping the large-scale distribution of matter and constraining cosmological parameters. However, its interpretation is affected by astrophysical systematics, especially intrinsic alignment, where galaxy shapes can become correlated through local gravitational and formation environments rather than lensing alone. If these additional correlations are not modelled properly, they may bias future weak-lensing measurements from surveys such as Euclid, Rubin Observatory LSST, and the Roman Space Telescope. This project investigates whether machine-learning methods can provide a compact representation of intrinsic-alignment two-point statistics. Instead of directly analysing high-dimensional theoretical power spectra, the project will train and evaluate dimensionality-reduction and generative models, such as PCA, autoencoders, and variational autoencoders, to compress these spectra into a small latent space while preserving their main physical dependence on cosmological and astrophysical parameters. The expected outcomes include a reproducible computational workflow, baseline compression models, quantitative reconstruction tests, latent-space interpretation, and a LaTeX research manuscript.

中文

弱引力透镜是研究宇宙大尺度物质分布并限制宇宙学参数的重要观测手段，但其解释会受到天体物理系统效应的影响。其中，内在对齐是一个关键例子：星系形状可能由于局部引力场和形成环境而产生相关性，而并非完全由引力透镜效应造成。如果这类额外相关信号没有被准确建模，未来的 Euclid、Rubin Observatory LSST、Roman Space Telescope 等巡天项目中的弱引力透镜测量可能会产生系统性偏差。本课题将探索机器学习方法能否为内在对齐两点统计提供一种更紧凑的表示方式。项目不会直接处

理高维理论功率谱，而是训练并评估 PCA、自编码器和变分自编码器等降维/生成模型，将功率谱压缩到少量潜在变量中，同时尽量保留其与宇宙学参数和天体物理参数相关的主要物理信息。预期成果包括可复现的计算流程、基线压缩模型、重构误差评估、潜变量空间解释，以及一篇 LaTeX 科研论文初稿。

课题创新点

1. 本课题将弱引力透镜中的天体物理系统误差建模转化为低维潜在表示学习问题，区别于直接在高维理论功率谱空间中进行建模。
2. 项目采用 conditional generative compression 的思路，将宇宙学/天体物理参数与功率谱压缩结合起来，探索 latent space 是否能够捕捉主要物理变化。
3. 预期结果可为未来 Euclid、Rubin、Roman 等巡天中的系统误差快速表示、模型比较和不确定性压缩提供概念验证。

预期研究成果

1. 建立 intrinsic-alignment two-point statistics / power spectra 的数据生成、读取、预处理与可视化流程。
2. 实现并比较 PCA、autoencoder、variational autoencoder 等低维压缩模型，评估 latent dimension 对重构质量的影响。
3. 分析 latent variables 与 cosmological / astrophysical parameters 之间的关系，判断压缩表示是否保留主要物理信息。
4. 完成可复现的 Jupyter Notebook、主要科研图表、LaTeX 论文初稿与项目展示材料。

科研计划

*TBA 为待确定具体时间

*项目推进过程中，研究进度可能会根据学生实际情况进行略微调整

科研项目 - 导师课程阶段
赛事相关辅导
比赛日程

科研项目 - 导师课程阶段-北京时间		
6/6(周六) 10:00-12:00	Class 01	Topic: Cosmology background Content: 1.Big Bang and cosmic expansion 2.Dark matter/dark energy 3.Large-scale structure Readings: NASA/ESA outreach materials on cosmology; selected project slides. Assignment: Write a 300-word summary of why cosmology needs observational probes.
		Topic: Weak lensing and intrinsic alignment Content: 1.Strong vs weak lensing 2.Galaxy-shape distortion 3.Intrinsic alignment as a systematic Readings: Bartelmann & Schneider, Weak Gravitational Lensing; Troxel & Ishak, Intrinsic Alignments review (selected sections). Assignment: Draw a concept diagram linking lensing, galaxy shapes, and IA contamination.
		Topic: From science question to ML task Content: 1.IA power spectra as high-dimensional vectors 2.Compression objective

		3.Reconstruction vs physical information
		Readings : Project abstract; Chollet, Deep Learning with Python, Ch. 1-2 (selected).
		Assignment : Draft research question, hypothesis, and expected outputs.
TBA	Class 04	Topic: Power spectra and toy data Content: 1.Multipole/scale intuition 2.Parameter-dependent curves 3.Toy spectra generation
		Readings : Introductory notes on power spectra and two-point statistics.
		Assignment: Generate and plot toy spectra; start literature-review notes.
TBA	Class 05	Topic: Python workflow and preprocessing Content: 1.NumPy arrays and shapes 2.Train/validation/test split 3.Normalisation/log scaling
		Readings : Chollet, Deep Learning with Python, Ch. 2 and Ch. 4 (selected).
		Assignment: Build a preprocessing notebook for toy or provided spectra.
TBA	Class 06	Topic: PCA baseline Content: 1.Dimensionality reduction 2.Explained variance 3.PCA reconstruction error
		Readings: Scikit-learn PCA documentation; short PCA tutorial.
		Assignment: Apply PCA with different component numbers and plot reconstruction residuals.

TBA	Class 07	Topic: Autoencoder compression Content: 1.Encoder/latent/decoder 2.Reconstruction loss 3.Latent dimension selection
		Readings : Chollet, Deep Learning with Python, neural-network and autoencoder notes.
		Assignment : Train a simple autoencoder and compare it with PCA.
TBA	Class 08	Topic: Model evaluation Content: 1.Loss curves 2.Overfitting/underfitting 3.Error metrics for spectra
		Readings: Chollet, Deep Learning with Python, Ch. 4 (selected).
		Assignment : Prepare Methods draft: data, preprocessing, PCA, autoencoder setup.
TBA	Class 09	Topic: Conditional compression Content: 1.Cosmological parameters 2.Astrophysical parameters 3.Conditioning strategy
		Readings : Project slide on conditional generative compression; selected IA modelling notes.
		Assignment: Design a conditional model diagram and update the methods section.
TBA	Class 10	Topic: Variational autoencoder concept Content: 1.Latent distribution 2.Sampling 3.Reconstruction term and KL regularisation

		Readings : Kingma & Welling, Auto-Encoding Variational Bayes (conceptual overview); Chollet Ch. 8 (selected).
		Assignment: Implement or review a simplified VAE pipeline.
TBA	Class 11	Topic: Physical interpretation of latent space Content: 1.Latent variables vs parameters 2.Sensitivity plots 3.Information preservation
		Readings: Selected examples of interpretable scientific ML.
		Assignment : Plot latent variables against input parameters and write preliminary observations.
TBA	Class 12	Topic: Main IA-spectra experiment Content: 1.Train final baselines 2.Compare PCA/AE/VAE 3.Select key figures
		Readings: Review previous notes and project slides.
		Assignment : Produce final result figures: spectra, residuals, error table, latent-space plot.
TBA	Class 13	Topic: Results and discussion Content: 1.Interpret reconstruction quality 2.Discuss model limits 3.Connect to future surveys
		Readings: Example astronomy/ML paper structure.
		Assignment: Draft Results and Discussion sections in LaTeX.
TBA	Class 14	Topic: Manuscript finalisation Content:

		1.Abstract/Introduction revision 2.Figure captions 3.Citation and LaTeX formatting
		Readings: Selected MNRAS/AAS paper-writing guidance.
		Assignment: Submit a complete manuscript draft for feedback.
TBA	Class 15	Topic: Final presentation and revision
		Content: 1.Final paper polish 2.Presentation storyline 3.Q&A preparation
		Readings: Review all project materials.
		Assignment: Submit final paper, slides, and presentation script.

2026 丘成桐中学科学奖-北美赛区

赛事官网: <https://www.yau-science-awards.org/>

July 1–September 15	Online registration and submission of application materials. The registration system will open on July 1.	Important Note: Students must be high school students during the registration period (July 1–September 15). If participating as a team, all members must be high school students from the same school at the time of registration. Cross-school teams are not allowed.
September 16–September 30	Review of submitted materials.	
October 1–November 2	赛事辅导	
October 1–November 2	Regional evaluations. Each region will select candidates to advance to the Global Final.	
November 3–November 10	The list of Global Finalists and their research reports will be publicly announced online. The final list of Global Final participants will then be confirmed.	
November 11–December 4	赛事辅导	
December 5–December 6	Global Final and Awards Ceremony.	The final competition will be conducted in English in the form of an oral defense, chaired by the International Judging Committee. Location: Tsinghua University.
December 6	Science Forum	

2025-2026 ISEF

---26 年时间线可供参考，27 年时间线更新后届时群内通知---

Aardvark Science Expo website:

<https://www.oes.edu/academics/academic-curriculum/science/aardvark-science-expo#:~:text=The%20Aardvark%20Science%20Exposition%20is,in%20tenth%20to%20twelfth%20grades.>

TBA	赛事辅导 (Aardvark Science Expo 赛事网站未查询到具体赛事信息，需要提前和学校老师了解今年校赛流程与要求以及对应时间节点)
2026/2/20	Aardvark Science Expo
TBA	赛事辅导
2026/4/6	Northwest Science Expo
TBA	赛事辅导
2026/5/9 - 2026/5/15	International Science and Engineering Fair (ISEF)